

TrES-3b

R-0363

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230619 · created 2026-07-18T14:05:49+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460114.6931 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.056
Transit depth (Rp/Rs)^2	3.72 +/- 2.17 [%]
Orbital Inclination (inc)	86.87 +/- 2.86
Airmass coefficient 1 (a1)	1.1 +/- 0.15
Airmass coefficient 2 (a2)	-0.09 +/- 0.31
Scatter in the residuals of the lightcurve fit is	13.58 %
Best Comparison Star	#3 - [90, 308]
Optimal Aperture	3.19
Optimal Annulus	9.68
Transit Duration (day)	0.0623 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.056 vs 0.16309 (deviation 0.53σ)
Duration fit vs published	0.0623 d vs 0.0567 d (deviation 0.92σ)
Mid-time O–C	2.7 min
Depth SNR	3.4
Residual scatter	13.58 %

Light curve

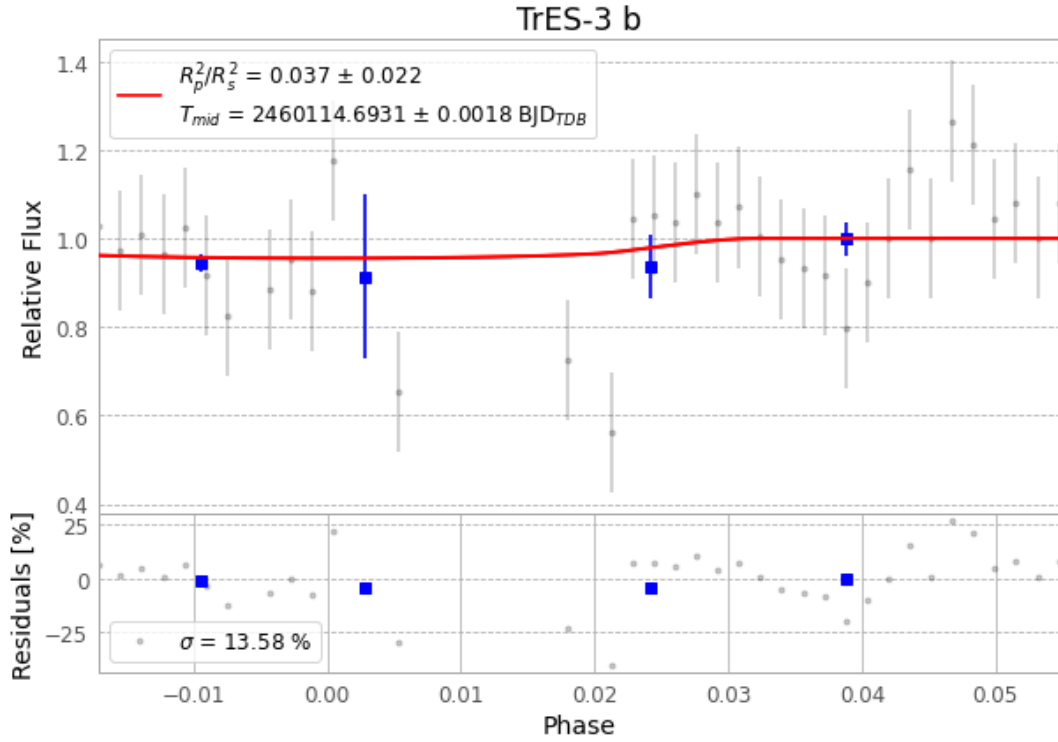


Figure 1 — FinalLightCurve_TrES-3 b_2023-06-18.png (SHA-256 319c2870660fda8c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 214], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2c09224eef124936...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9e597cf

Data provenance

46 FITS frames (30 MB), TRES-3230619040016.FITS → TRES-3230619061515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230619.log (7a06d6394b31...), FinalLightCurve_TrES-3 b_2023-06-18.png (319c2870660f...), FinalLightCurve_TrES-3 b_2023-06-18.csv (aa942268c2eb...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 14:05Z.